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Juniper: a 3D plotting framewrork for effective multi-dimensional scatterplots

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Software

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Summary

Computationally and data intensive workloads including design space exploration or large studies often lead to multi-dimensional results, which are often not trivial to digest with conventional plotting software. 3D scatterplots can be a powerful technique to visualise and explore such datasets, especially with the help of colour mapping and other approaches to represent more than the 3 Cartesian dimensions. However, modern software commonly lacks this multi-dimensional functionality or is ineffective. One such limitation is the frequent use of isometric axes that produces same-sized points, which is equivalent to removing one entire dimension.

In this work, novel open-source JavaFX-based plotting framework is presented that focuses on easy exploration of multi-dimensional datasets, and provides unique features or feature combinations to improve knowledge transfer from single stand-alone plots.

Juniper includes a list of *unique visual cues* that can be organised according to the corresponding aim:

- *Depth perception*: depth-based colour mapping (traditionally only for an axis), fog emulation, a highly adjustable field of view (FoV).
- *Readability of the plotted points and labels in the 3D space*: multi-axis light paths (alternative to direction drop lines), light paths as 3D objects shaped according to the points (instead of same-sized overlaid lines), 2D projections corresponding to the point shape and colour (not disassociated), 2D surfaces as grids, labels as 3D objects (meshes (F(X)yz, 2013) instead of overlays, while still facing the camera).
- *Exploration of multiple dimensions*: high number of possible dimensions organised in an association panel, automatic discrete data support such as for strings, automated point label decluttering such as by applying a top-N selection.

Statement of need

Juniper is targetted towards visualisation research, such as on the exploration of the aforementioned unique visual cues. Ongoing research that uses this framework includes an empirical study exploring the effectiveness of these cues on 3D scatterplots as static 2D images (Papaphilippou & Hederman, 2024). As it is open-source and highly-extensible such as by having the entirety of the plot in the 3D space, Juniper can accelerate future research on other unconventional scaterplotting techniques.

As an advanced 3D scatterplotting tool, Juniper can also help with research explorations that rely on knowledge extraction from complex data. With its innovative features, it focuses on quickly and effectively expressing multiple-dimensions with realistic 3D scatterplots. In

combination with its graphical user interface as a desktop application, it can provide early insights from experimental data with high dimensionality including computer simulations.

While a basic 3D scatterplot functionality can be included in conventional software such as office suites, there are limitations for specialised applications. As described in the summary, these relate to depth information communication, user interactivity, and the ability to represent multiple dimensions. More feature-rich 3D scatterplots are usually proprietary and closed-source, further restricting their applicability to visualisation research.

Use case example

This section presents an example demonstrating how Juniper can quickly generate insightful stand-alone plots from multi-dimensional data by using a subset of the unique features. The explored dataset is a list of the top 500 supercomputers in the world, which incorporates a comprehensive set of attributes for each supercomputer in a spreadsheet format (TOP500.org, 2024).

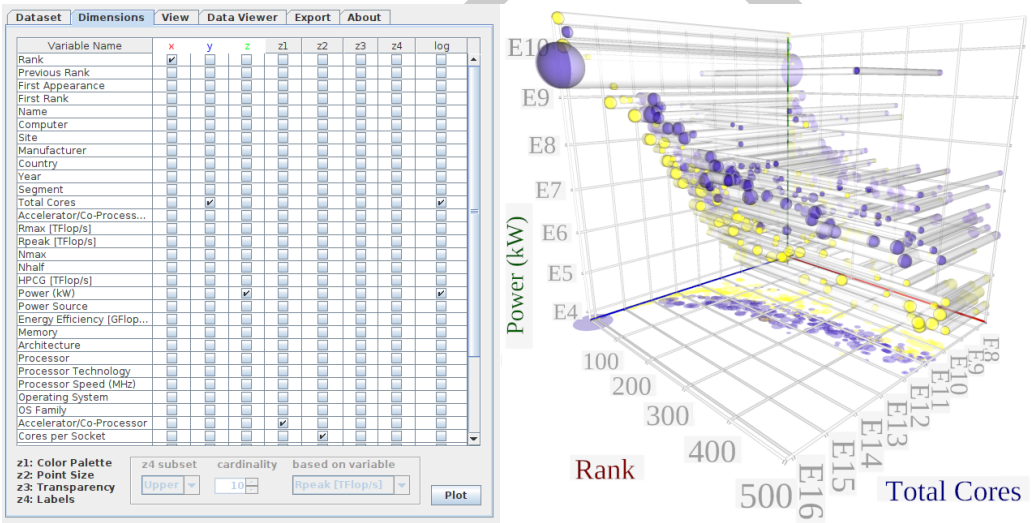

Figure 1: Assigning columns to dimensions from a *.csv*-like dataset on supercomputers for a 5D plot

The resulting plot is shown in figure 1, right. Based on the dimension assignment, as shown in figure 1 left, the *x*-axis (red) is the rank of the supercomputer entry. The *y*-axis (blue) is the total number of CPU (central processing unit) cores in the supercomputer, and the *z*-axis is the total power in kilowatt-hours. The latter-two have their log checkbox checked for applying a logarithmic scale, due to the high variability in the orders of magnitudes for those values.

There are numerous insights that can be derived by looking at the five-dimensional plot in figure 1, right. First, if we ignore the 4th and 5th dimensions (point size and colour), there seems to be a high correlation between the rank and the total number of cores and power, when we consider them in pairs. If we take the projections on the *xy* surface (the floor), the more performant (faster) the supercomputer is (rank), the more likely it is to have a higher number of cores. Similarly, the higher the core count, the more likely it is to consume more energy. Additionally, the faster the supercomputer, the more energy it consumes.

When we take the colour mapping into account (*z1*, "Accelerator/Co-processor"), we can see that the GPU-equipped supercomputers (yellow) consume less power than CPU-only supercomputers (purple) for the equivalent performance. With the point-size dimension (*z2*, "Cores per Socket") we can see that the GPU-equipped supercomputers generally feature smaller CPUs than CPU-only supercomputers. This makes sense due to the fact that CPU-only

supercomputers migrate some of the GPU functionality to the CPU sockets, such as by having a high core count (notably the higher ranking ones, which are mostly more recent). The largest sphere in the plot is Sunway TaihuLight in China, the 13th supercomputer in this edition of the list, whose manycore CPUs combine 260 cores in a single socket (Fu et al., 2016).

Related work

The presented visual cue list is unique to the best of our knowledge, though keeping intuitiveness in mind, it could be possible to find resemblance in related work, or be able to reproduce them with additional scripting. Hence, the contribution in this aspect is at least their coexistence in Juniper, while being open-source and research-friendly.

A close framework in regard to the explored functionality is OriginPro (OriginLab Corp., 2024). It supports a superset of plot types, but its 3D scatterplots currently lack the aforementioned unique features. Underneath, a mix of 2D raster renderings and vector components are used. This creates an impressive clarity effect in printouts (axis labels in a text format), but also results in some loss of physical properties and interactive navigation flexibility. This includes quantised ranges for rotation and FoV adjustment, and some pixelation when zooming in. Being a more general and proprietary analytics framework, this is reflected in the license cost at 645 USD for a node-locked annual subscription.

With respect to the dimension assignment facility with checkboxes, it exists in certain software with a limited dimensionality. One instance is "DB Browser for SQLite", an open source database management system focusing on local databases (M Piacentini et. al, 2024). It also operates on tabular data, though its functionality is restricted to 2D plots. ParaView (Ahrens et al., 2005) is an advanced visualisation framework that also provides point-and-click functionality for the creation of plots. For more advanced figures such as 3D scatterplots with point projections, additional coding is required with a scripting language.

Other examples of software for generating advanced 3D scatterplots include *pygmt.Figure.plot3d* (Uieda et al., 2020) and the research-informed ParaProf (Bell et al., 2003) that also uses Java. Both are relatively simplistic when compared to the presented framework, and lack the discussed depth perception and label legibility functionality. Another scatterplot solution that uses JavaFX 3D and the FXyz3D component library is within the deep space trajectory explorer (Phillips et al., 2019), which specialises on multibody trajectory analysis and is not publicly available.

Basic point projections on the back of the view box is an existing approach for improving plot readability. One example is a WebGL online 3D scatterplot viewer (Hergott, 2013) that features point projections, as well as drop lines when the user hovers the mouse cursor over a point. When compared to the presented software, it is mostly an interactive proof-of-concept, with same-sized points and labels lacking an automatic heading adjustment.

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